



MOTIVATION

Existing methods struggle with
semantic alignment & **physical plausibility**.

Previous Methods

- ✗ Semantic mismatch
- ✗ Unstable contact



Ours (AffordGrasp)

- ✓ Semantic aligned
- ✓ Physically plausible



KEY CONTRIBUTIONS

- 1 A novel framework that leverages language-driven affordances to bridge semantics and geometry.
- 2 An automatic annotation engine for scalable dataset enrichment with instructions & affordances.
- 3 A physics-aware diffusion model with a two-stage training strategy for efficient inference.



EVALUATION

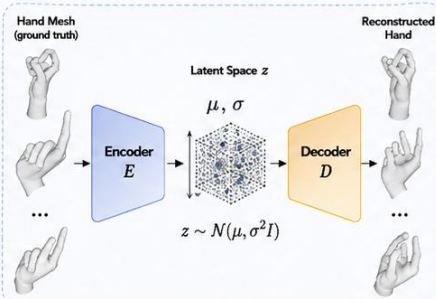
- 1 **Physical Plausibility Assessment**
Evaluated by hand-object penetration volume (mm^3 , ↓) and contact ratio (% ↑).
- 2 **Grasp Stability Assessment**
Measured by the gravitational displacement (mm , ↓) of the object's center of mass in simulation.
- 3 **Diversity Assessment**
Assessed using entropy of cluster assignments (↑) and average cluster size (↑) based on 3D hand keypoints.
- 4 **Semantic Accuracy Assessment (ACC)**
Measured by the classification accuracy (%) of generated hand poses matching the instruction's affordance class.

METHOD OVERVIEW

3 Three-stage Training Strategy

Stage 1: VAE Training (Hand Reconstruction)

Learn a VAE to encode hand meshes to a latent space and decode them back.

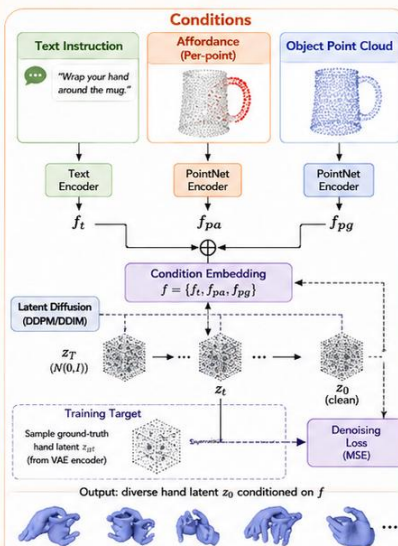


Reconstruction Loss

$$\text{Geometry Loss (Chamfer)} + \text{Normal Loss} + \text{MANO Param Loss}$$

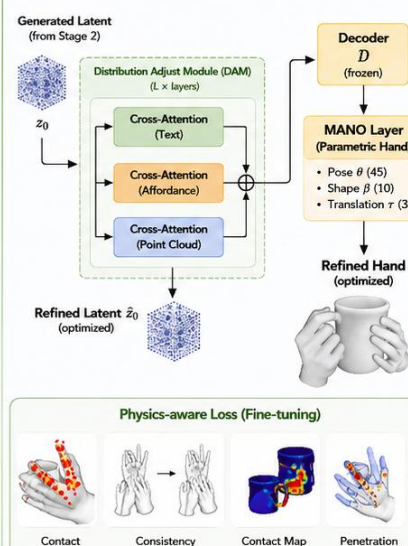
Stage 2: Latent Diffusion Training (Conditioned Generation)

Train a latent diffusion model to generate diverse hand latent representations conditioned on text, affordance, and object point cloud.

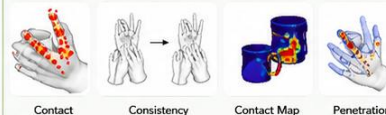


Stage 3: DAM Fine-tuning (Post-optimization)

Use DAM to refine the generated latent for better physical plausibility and contact, then decode to MANO hand.



Physics-aware Loss (Fine-tuning)



Training Flow



Inference Pipeline (Test Time)



Visualization

